

Lloyds Banking Group

Data Science & Analytics

Customer Retention Enhancement

Task 1: Data Gathering & Exploratory Analysis

SmartBank (subsidiary of Lloyds Banking Group)

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Executive Summary

SmartBank has observed a rise in customer attrition among working professionals and small-business owners. This report covers Phase 1 of the Customer Retention Enhancement project: data gathering, exploratory data analysis, and preparation of the dataset for predictive modelling. The work draws on five sheets covering 1,000 customers, merged into a single customer-level table with 26 predictor features built from 15 engineered features (11 from transactions, 3 from customer service, 1 from online activity) plus the encoded demographic and channel variables.

The customer base in this sample shows a 20.4% churn rate so moderately imbalanced and a non-trivial classification problem. Login frequency has the strongest individual linear association with churn in the numerical feature set, but the correlation is only -0.082, which explains less than 1% of the variation in churn. Differences in churn rate by service channel, preferred product category, marital status and income level are visible in the data but none is statistically significant at the 0.05 level after chi-square testing. The five SpendShare features differ between churned and retained customers by less than three percentage points in every category. These should be treated as candidate hypotheses for the modelling phase instead of findings to act on now.

A regularised logistic regression baseline scored ROC-AUC of 0.514 and PR-AUC of 0.241 in 5-fold cross-validation. This is barely better than chance, and is consistent with the weak individual signals seen in the EDA. Phase 2 should expect tree-based models to do somewhat better through interactions, but the realistic ceiling on this dataset is modest.

Three issues affect what Phase 2 can credibly conclude. First, the transaction window (2022) and online-activity window (2023) do not have a defined relationship to the churn-measurement date, so features like LoginFrequency may be downstream of churn rather than predictive of it. Second, the dataset has no tenure variable, no occupation field, and no SME-account flag, which means the brief's target segments cannot be isolated in the data as supplied. Third, with only 204 positive cases, feature-importance estimates and held-out scores from any model will carry wide confidence intervals.

The deliverable is a single-sheet cleaned dataset of 1,000 rows by 28 columns (1 ID + 1 target + 26 predictors), with a data dictionary and Phase 2 modelling notes embedded in the workbook. The dataset is unscaled by design meaning the scaling must be fitted on the training fold inside a cross-validation pipeline, not on the full dataset upfront.

1. Data Sources and Selection

The provided workbook contains five sheets. All five are used. Coverage checks were performed against CustomerID before any aggregation.

Source	Rows	Unique customers	Role
Customer_Demographics	1,000	1,000	Age, gender, marital status, income level.
Transaction_History	5,054	1,000	Spend volume, value, recency, product mix.

Source	Rows	Unique customers	Role
Customer_Service	1,002	668	Complaint and resolution signals (informative absence for 332 customers).
Online_Activity	1,000	1,000	Digital channel and login behaviour.
Churn_Status	1,000	1,000	Binary target label.

Table 1. Source sheets and their role in the analysis.

2. Methodology

The workflow follows the standard sequence: data quality checks, aggregation to one row per customer, feature engineering, statistical analysis with confidence intervals and proper tests, encoding, then dataset export. Python 3 was used with pandas, NumPy, Matplotlib, Seaborn, SciPy and scikit-learn.

2.1 Quality checks

- Missing values: none across any sheet.
- Duplicate rows: none.
- Range checks: Age 18 to 69, AmountSpent £5.18 to £499.86, LoginFrequency 1 to 49. All within plausible bounds.
- Outliers: tail values in AvgSpent, MaxSpent and MinSpent are not data-entry errors. They sit at the natural boundaries of fairly uniform distributions and have been retained.

2.2 Statistical tests applied

- Wilson 95% confidence intervals for category-level churn rates (better than the normal-approximation interval at small cell counts).
- Pearson chi-square tests of independence for each categorical feature against churn. The same test (no Yates continuity correction) is used in both the figure titles and the report text.
- Cramér's V to quantify the effect size of categorical associations.
- Point-biserial correlations with Fisher-z 95% confidence intervals for numerical features against the binary target.
- Mann-Whitney U for the bounded spend-share distributions, which are non-normal.
- Variance inflation factors (VIF) computed via R-squared of each numerical feature regressed on the others, to quantify multicollinearity.

3. Exploratory Findings

3.1 The target variable

204 of 1,000 customers churned (20.4%). The 4:1 ratio of retained to churned customers is shown in Figure 1. This imbalance shapes the choice of evaluation metric in Phase 2 as accuracy alone is inadequate, since a no-churn classifier would score 79.6%.

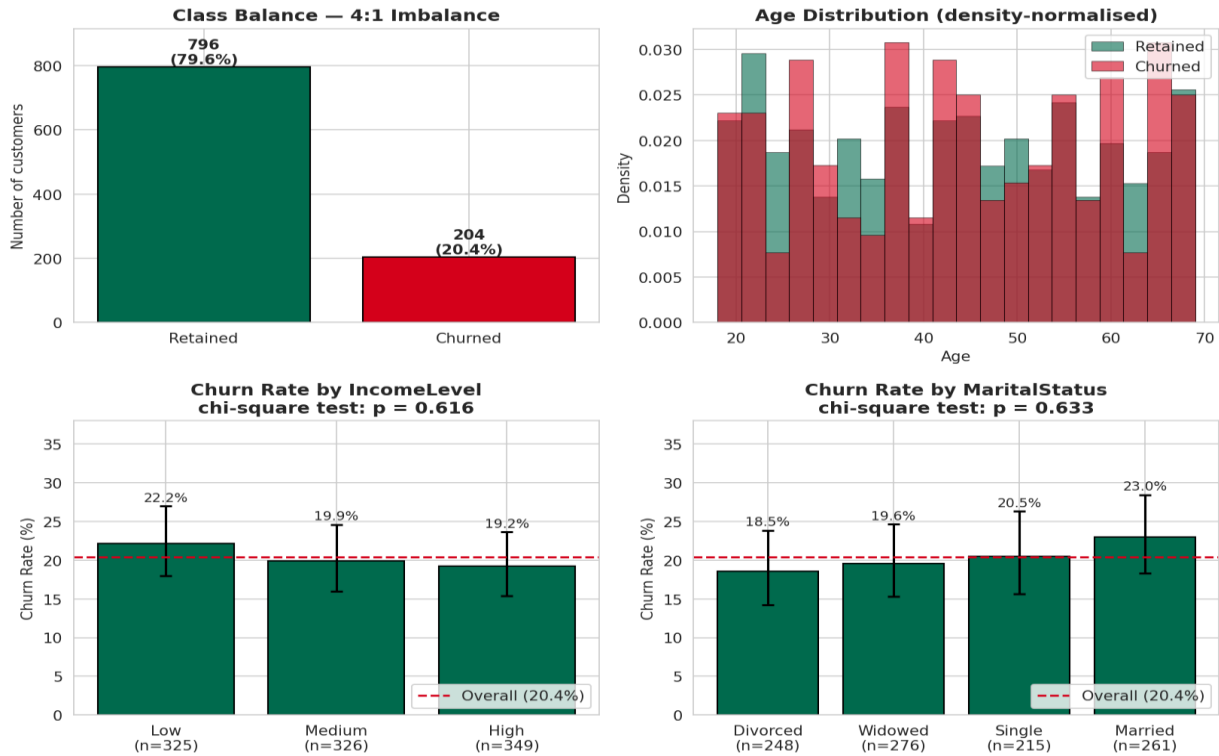


Figure 1. Class balance and demographic breakdowns. Error bars are Wilson 95% confidence intervals; chi-square p-values are shown above the bottom-row charts. None of the demographic differences crosses the 0.05 significance threshold.

3.2 Demographic associations are not significant

Group churn rates differ slightly. Low-income customers churn at 22.2% vs 19.2% for high income. Married customers churn at 23.0% vs 18.5% for divorced. But the 95% confidence intervals overlap substantially in every case, and the chi-square tests give p-values of 0.616, 0.633 and 0.621 for income, marital status and gender respectively. Cramér's V values are all under 0.05, which means the effect sizes are negligible. Any retention action targeted at these groups would not be evidence-based.

3.3 Behavioural patterns

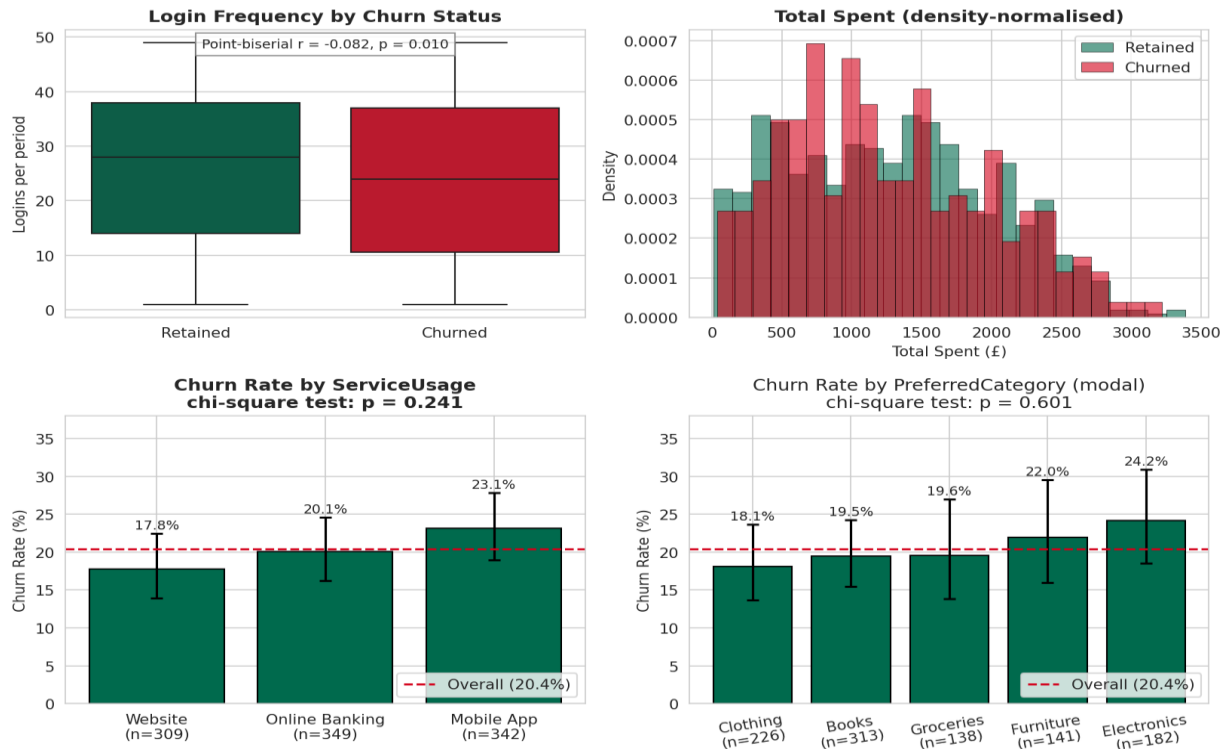


Figure 2. Behavioural patterns. The two distributions overlap heavily. Service channel and preferred category show visible differences but the chi-square p-values exceed 0.24 - these are candidate hypotheses, not findings.

Login frequency has the strongest individual numerical association with churn. The point-biserial correlation is $r = -0.082$ with 95% CI -0.143 to -0.020 and $p = 0.010$. The direction is intuitive - churners log in less often but the effect is small. Login frequency explains under 1% of churn variation on its own, so it should be tested as part of a multivariate model rather than treated as a standalone churn driver.

The chi-square tests in Figure 2 give a global view: $p = 0.241$ for ServiceUsage and $p = 0.601$ for the modal preferred category. To contextualise these, I also ran exploratory unadjusted two-proportion z-tests on the highest-versus-lowest pairs within each variable; $p = 0.095$ for Mobile App vs Website and $p = 0.136$ for Electronics vs Clothing. The chi-square tests cover the full categorical variable; the pairwise comparisons are unadjusted exploratory checks. Neither view crosses 0.05, so the patterns are hypotheses to test in a multivariate model and are not the findings to act on.

3.4 Spend-share patterns

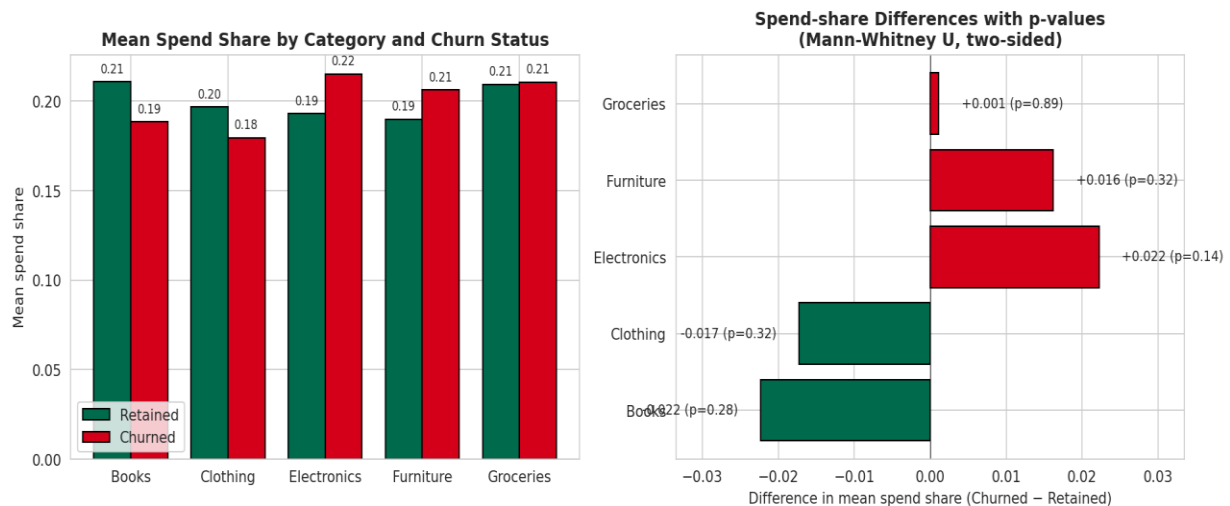


Figure 3. Mean spend share by category and churn status, with the difference and Mann-Whitney U p-value per category. Churned customers spend a slightly higher share on Electronics and Furniture; retained customers spend a slightly higher share on Books and Clothing. None of the differences is statistically significant.

The five spend-share features preserve the customer's spending mix instead of collapsing it to a single label. Differences between retained and churned customers are small in every category under three percentage points and the Mann-Whitney U p-values range from 0.14 (Electronics) to 0.89 (Groceries). The direction is consistent with the modal-category view in Figure 2 (Electronics churn is higher; Clothing churn is lower), but the effect sizes show that spend mix carries no strong individual signal. The shares are kept in the final dataset so a multivariate model can pick up any interactions with other features.

3.5 Service-friction signals

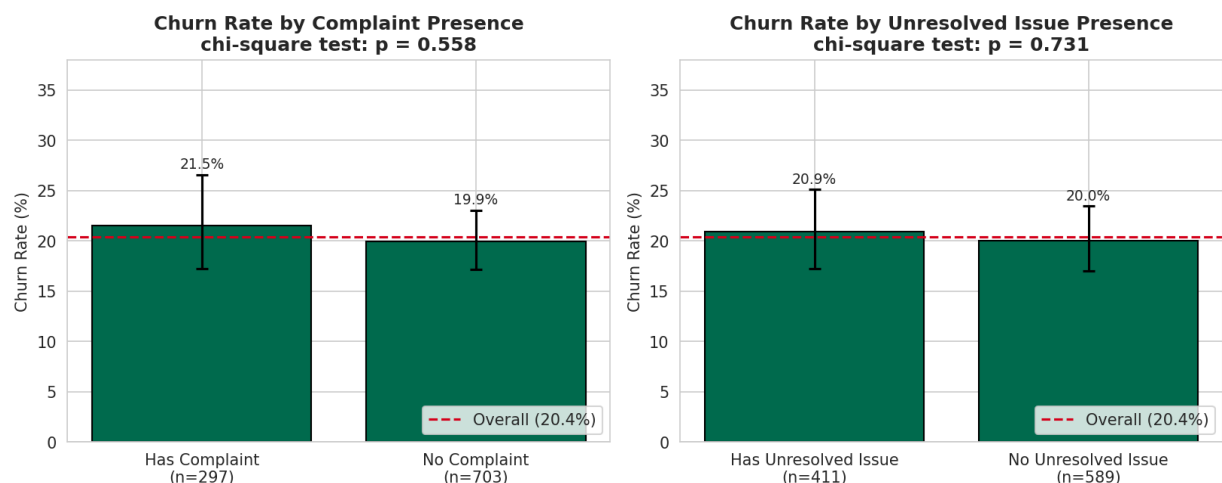


Figure 4. Churn rates for customers with versus without logged complaints and unresolved issues. Wide confidence intervals reflect the small sample of contacting customers.

Customers with at least one complaint churn at 22.1% versus 20.0% for those without. The chi-square test (no Yates correction) gives $p = 0.558$. For unresolved issues, $p = 0.731$. The dataset caps service interactions at two per customer, which limits the discriminative power any service-related feature can offer. Real banking data would have a longer interaction history.

3.6 Multicollinearity is severe among transaction features

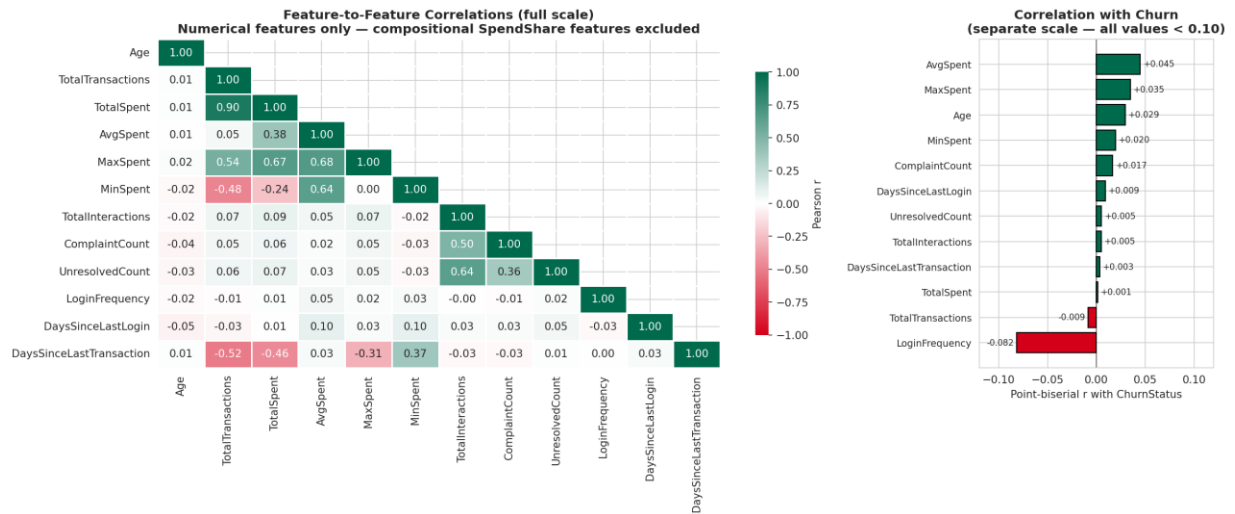


Figure 5. Left: full-scale feature-to-feature correlations for numerical features. Right: separate-scale point-biserial correlations with churn. The two panels are deliberately on different scales because feature-to-feature correlations reach 0.90 while target correlations stay below 0.10.

The full-scale heatmap on the left tells a different story from a churn-only view. TotalTransactions and TotalSpent are correlated at $r = 0.90$ - TotalSpent is essentially TotalTransactions multiplied by AvgSpent, so the relationship is built in. AvgSpent and MaxSpent overlap at $r = 0.68$; TotalInteractions and UnresolvedCount at $r = 0.64$. Variance inflation factors confirm this:

Feature	VIF	Interpretation
TotalSpent	18.4	Severe - drop or combine for linear models
AvgSpent	17.0	Severe - drop or combine for linear models
TotalTransactions	16.9	Severe - drop or combine for linear models
MaxSpent	6.8	Moderate - caution for linear models
MinSpent	5.9	Moderate - caution for linear models
TotalInteractions	2.0	Acceptable
Other features	< 2.0	Acceptable

Table 2. Variance inflation factors above 5 indicate problematic multicollinearity for linear models. Note: the five SpendShare_<Category> features are excluded from this table because they are compositional (sum to 1.0 per customer). One of them must be dropped before any non-regularised linear model regardless of what the VIF would read.

For tree-based models, multicollinearity does not break the fitting process but it does scramble feature-importance rankings when two features carry the same signal, the importance is split arbitrarily between them. For linear models, regularisation (L1 or L2) is needed, or features must be dropped manually.

3.7 Baseline model - concrete evidence the signal is weak

To check the EDA findings end-to-end, I fitted a regularised logistic regression with class-balanced weighting inside a scikit-learn pipeline (scaler fitted on the training fold only). The 5-fold cross-validation results on the training set were ROC-AUC 0.514 ± 0.040 and PR-AUC 0.241 ± 0.026 . The 0.241 PR-AUC is essentially at the no-information floor (the positive-class prevalence of 0.204). This confirms quantitatively what the EDA suggests qualitatively, in their current form, the supplied features do not separate churners from retained customers well. Phase 2 should expect to do somewhat better with non-linear models, but should treat any score above 0.80 ROC-AUC as a red flag for leakage rather than a success.

4. Limitations and Modelling Risks

This section sets out the four most serious caveats around the work. Each affects what Phase 2 can credibly claim. They should be discussed with the project sponsor before model development starts.

4.1 Target leakage risk via the churn-measurement date

Highest-priority risk

Transaction dates run through 2022. Online-activity dates run through 2023. The churn label has no associated measurement date. If a customer churned during 2022, their low 2023 login frequency is a consequence of leaving rather than a predictor of it. A model trained on this data may learn to "predict the past" rather than predict future churn. Before Phase 2, the team must verify when ChurnStatus was measured and define an observation window for features that ends strictly before that date.

Two features are particularly exposed: LoginFrequency and DaysSinceLastLogin (both measured in 2023, after the transaction window). If these were measured post-churn for some customers, their predictive value is artificial. The cleaned dataset flags this in the data dictionary so Phase 2 cannot use them without first checking. I also note that the reference date for the recency features (1 January 2024) is provisional. It is set to one day after the latest observed login date for tidy arithmetic, but the true prediction snapshot may be earlier or later. If the transaction and online tables were extracted over different observation windows, the recency features should be rebuilt after the windows are aligned with the project sponsor.

4.2 The brief's target segments are not in the data

The project brief identifies working professionals and small-business owners as the at-risk segments. The supplied data contains no occupation, employment status, business-account flag, SME marker, salary-payment pattern, or relationship-manager indicator. With only age, gender, marital status and income level on the demographic side, the data cannot directly answer the brief's question. Phase 2 results should be framed as a general-population churn model rather than a working-professional or SME-owner model until appropriate features are added.

4.3 Missing tenure

Customer tenure (how long the customer has held an account or relationship with the bank) is often an important churn predictor in retail banking, and would be a strong candidate for inclusion. The supplied data contains no such variable. Recommendation: request account-opening date, product-holding duration and relationship-depth fields in the next data refresh.

4.4 Sample size and synthetic-data caveat

Only 204 customers in the dataset are churners. After a typical 80/20 train/test split, roughly 40 churners would sit in the test set as too few for stable performance estimates. Cross-validation will be necessary, and confidence intervals on every reported metric should be wide.

The data itself has several hallmarks of a synthetic or training sample as no missing values across 8,000+ records, perfectly uniform-looking distributions, transaction counts capped at nine and service interactions capped at two. Real banking data is not this clean. Findings from this analysis are methodological practice and should not be treated as actionable market intelligence for SmartBank without validation on a real dataset.

5. Data Cleaning and Feature Engineering

5.1 Aggregation

Transaction and customer-service tables were aggregated to one row per customer. The 332 customers absent from the service table were given zero values, since no contact is itself an informative signal of zero friction.

5.2 Engineered features (15 in total)

The final dataset contains 26 predictor features. Of these, 15 are engineered customer-level features:

- From transactions (11 features): TotalTransactions, TotalSpent, AvgSpent, MaxSpent, MinSpent, DaysSinceLastTransaction, and the five SpendShare_<Category> compositional features (one per category: Books, Clothing, Electronics, Furniture, Groceries).
- From customer service (3 features): TotalInteractions, ComplaintCount, UnresolvedCount.
- From online activity (1 feature): DaysSinceLastLogin (derived from LastLoginDate against the reference date).

The remaining 11 predictor features come from encoded demographic and online variables: Age (raw), Gender_M (binary), IncomeLevel_Ord (ordinal), four Marital_* dummies, three Channel_* dummies and LoginFrequency (raw).

5.3 Encoding

- IncomeLevel - ordinal (Low=0, Medium=1, High=2) because the categories have a natural order.
- Gender - binary (Gender_M).

- MaritalStatus and ServiceUsage - one-hot with all categories kept. For non-regularised linear models, drop one column per block (see Modelling_Notes in the workbook).
- PreferredCategory replaced by the five SpendShare_* compositional features. These sum to 1.0 per customer, so one should be dropped for a non-regularised linear model.

5.4 Scaling - handled in Phase 2, not here

Data-leakage avoidance

I have not fitted a scaler on this dataset. Fitting StandardScaler on all 1,000 rows before splitting into train and test sets causes train-test contamination as the test fold's mean and standard deviation enter the training preprocessing. This is data leakage in the broad sense, and is distinct from the target-leakage risk in section 4.1, which concerns the churn-measurement date. The deliverable workbook ships only the unscaled data. Scaling must be fitted on the training fold inside a scikit-learn Pipeline. The Modelling_Notes sheet of the workbook gives the exact pattern.

5.5 Final dataset shape

1,000 rows by 28 columns: 1 CustomerID, 1 ChurnStatus target, and 26 predictor features. The class balance (79.6% / 20.4%) is preserved as observed and will be handled at model-training time, not earlier.

6. Recommendations for Phase 2

6.1 Before building any model

- Verify when ChurnStatus was measured and trim any feature whose measurement window extends past that point. This is the single most important step.
- Request occupation, employment status, and SME-account flags from the source system so the brief's target segments can be properly studied.
- Request tenure and account-opening dates.

6.2 Modelling approach

- Start with simple, regularised baselines: logistic regression with L1 or L2 regularisation, and a shallow decision tree. These give a defensible floor before any complex model is built. The baseline in this report (ROC-AUC 0.514, PR-AUC 0.241) is a starting reference.
- Compare against tree-based ensembles with limited depth and careful hyperparameter tuning (random forest, gradient boosting). Do not assume XGBoost or LightGBM with default settings will improve on a regularised baseline at this sample size.
- Use a proportional train/test split on ChurnStatus to preserve the 79.6 / 20.4 class balance in both folds. Hold out 20% for a final test set; cross-validate on the remainder.

- If using SMOTE or other resampling, apply it inside the cross-validation folds but never before splitting, which would leak the test distribution.

6.3 Evaluation

- Primary metrics: precision, recall and F1 for the positive (churn) class, ROC-AUC and PR-AUC. PR-AUC is preferred over ROC-AUC at this class imbalance.
- Recall at top-k: if the retention team can only contact, say, the top 10% riskiest customers, report `recall@10%` as a business-aligned metric.
- Calibration: check the reliability diagram and Brier score. Retention spend depends on actual probability levels, not just on rankings.
- Threshold tuning: set the classification threshold by business cost i.e. the cost of a missed churning vs the cost of contacting a loyal customer who would not have left.
- Report confidence intervals on every headline metric. With 204 positives, a single point estimate is misleading.

6.4 Feature interpretation

- Use SHAP values for the chosen model rather than raw feature-importance rankings. With strong multicollinearity in the transaction features, importance rankings on their own can be misleading.
- Sanity-check whether the model leans on potentially leaky features (`LoginFrequency`, `DaysSinceLastLogin`). If it does, refit without them and compare.

7. Responsible Use

The dataset contains gender, marital status and income level. In a UK banking context these are sensitive characteristics as gender is protected under the Equality Act 2010, and using marital status or income as a retention-targeting variable carries reputational risk even where it is legally permissible. Phase 2 should:

- Report performance separately for each gender, income band and marital-status group. Large gaps in recall or precision across groups must be investigated before any model is deployed.
- Compare a model fitted with these features against one fitted without them. If the predictive value of the protected characteristics is small (which the EDA suggests), the model without them may be the safer choice.
- Document why each feature is used and what retention action it might drive. A model used purely for proactive outreach has a lower fairness risk than one used to gate pricing or product access.
- Treat any geographic, occupational or business-account features added later under the same review.

Final feature use should be reviewed with legal, compliance and model-risk stakeholders before any production deployment. The points above set a minimum analytical bar; they do not substitute for that review.

8. Deliverables

- `Cleaned_Customer_Churn_Data.xlsx` - three sheets: `Cleaned_Data` (1,000 × 28), `Data_Dictionary` (19 feature entries with notes on multicollinearity and leakage risk), and `Modelling_Notes` (13 numbered guidelines for Phase 2).
- `Task1_EDA_Report.docx` - this report.
- `Task1_EDA_Notebook.ipynb` - the analysis notebook, which reproduces every figure, table and statistical test in the report end-to-end.